



LABORATORIO DI INTELLIGENZA ARTIFICIALE APPLICATA

A.K.A. **AI APPLICATIONS
IN PRACTICE**

11b - Statistics Prerequisites

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Random Variables & Probability

Discrete

A random variable X maps outcomes to numbers.

Probability Mass Function (PMF):

$P(X = x)$ for each possible value x

Example — coin flip:

$X \in \{0, 1\}$, $P(X = 1) = p$, $P(X = 0) = 1 - p$

Normalization:

$$\sum_i P(X = x_i) = 1$$

Continuous

X takes values in a continuous range.

Probability Density Function (PDF):

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Cumulative Distribution (CDF):

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

Normalization:

$$\int_{-\infty}^{+\infty} f(x) dx = 1, \quad f(x) \geq 0$$

In ML: input features are random variables; labels are random variables we want to predict.

Key Probability Distributions

Bernoulli

$$P(X = 1) = p, \quad P(X = 0) = 1 - p$$

Single binary trial. Used in: binary classification labels.

Binomial

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Number of successes in n trials. Used in: counting positive outcomes.

Gaussian (Normal)

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The "bell curve", defined by μ and σ .
Used in: normalization, noise, Bayesian priors.

Uniform

$$f(x) = \frac{1}{b-a} \quad \text{for } x \in [a, b]$$

Equal probability everywhere.
Used in: random initialization, sampling.

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Expectation, Variance & Standard Deviation

Expected Value (Mean)

$$\mathbb{E}[X] = \sum_i x_i P(X = x_i) (\text{discrete})$$

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f(x) dx (\text{continuous})$$

The "center of mass" of the distribution. Linearity: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$

Variance

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Measures spread around the mean. Scaling: $\text{Var}(aX + b) = a^2 \text{Var}(X)$

Standard Deviation

$$\sigma = \sqrt{\text{Var}(X)}$$

Same units as X. For a Gaussian: 68% within $\pm 1\sigma$, 95% within $\pm 2\sigma$, 99.7% within $\pm 3\sigma$.

Bias-variance: $\mathbb{E}[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Var} + \sigma^2$

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Conditional Probability & Bayes' Theorem

Conditional Probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

The probability of A given that B has occurred.

Independence: $P(A | B) = P(A)$
then A and B are independent.

Bayes' Theorem

$$P(A | B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}$$

Update beliefs based on new data.

Why This Matters in ML

Supervised learning models $P(Y | X)$: the probability of label Y given features X.

$$P(\text{spam} | \text{email features})$$

Logistic regression models $P(Y = 1 | X)$ via the sigmoid function.
Bayesian inference and Naive Bayes classifiers directly apply Bayes' theorem.

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Covariance & Correlation

Covariance

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Measures the direction of the linear relationship between two variables.

Cov > 0: both increase together | Cov < 0: inverse | Cov = 0: no linear relation.

Pearson Correlation Coefficient

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y} \in [-1, +1]$$

Normalized covariance — scale-independent measure of linear dependence.

$\rho = +1$ perfect positive | $\rho = -1$ perfect negative | $\rho = 0$ uncorrelated (not necessarily independent!).

In ML: feature correlation analysis (multicollinearity), covariance matrices in PCA and Gaussian models, standardization (zero mean, unit variance) before gradient descent.

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